

Part I

Linear Constant-Coefficient Differential Equations

An N th-order LCCDE is given by

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

1 Solution

The solution to an LCCDE consists of two parts: a *homogeneous solution* plus a *particular solution*.

$$y(t) = y_h(t) + y_p(t)$$

1.1 Homogeneous Solution & Natural Response

The homogeneous solution is the solution to the homogeneous equation

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = 0$$

It has N constants to be determined by the N auxiliary conditions.

Characteristic Polynomial

$$P(\lambda) := \sum_{k=0}^N a_k \lambda^k$$

Characteristic Equation

$$P(\lambda) = \sum_{k=0}^N a_k \lambda^k = 0$$

If $P(\lambda) = 0$ has N distinct complex roots, $\lambda_1, \lambda_2, \dots, \lambda_N$, then the solution is

$$y_h(t) = \sum_{k=1}^N C_k e^{\lambda_k t}$$

where $C_1, C_2, \dots, C_N$ are constants to be determined.

If $P(\lambda) = \prod_{k=1}^n (\lambda - \lambda_k)^{r_k}$, where $\lambda_1, \lambda_2, \dots, \lambda_n$ are distinct roots of $P(\lambda) = 0$, then the solution is

$$y_h(t) = \sum_{k=1}^n (C_{k,r_k-1} t^{r_k-1} + \dots + C_{k,1} t + C_{k,0}) e^{\lambda_k t}$$

where $C_{1,0}, \dots, C_{n,r_n-1}$ are N constants to be determined.

1.2 Particular Solution & Forced Response

Part II

Linear Constant-Coefficient Difference Equations